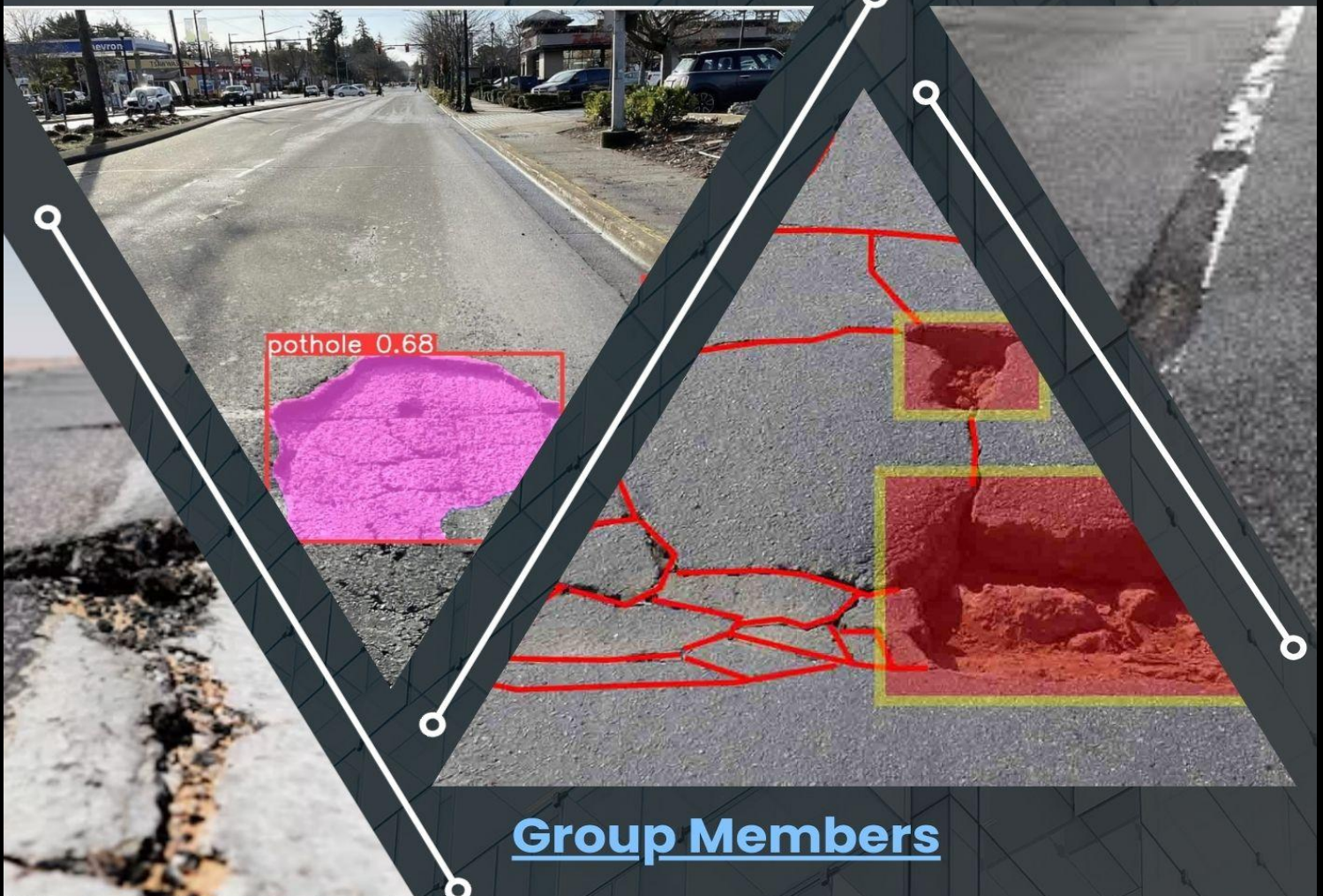




AUTOMATED ROAD CONDITION MONITORING SYSTEM



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INTRODUCTION

Currently, transportation related changes have become increasingly noticeable across Sri Lanka. Not having a formal transportation system has been a major issue in Sri Lanka nowadays. Due to this, the number of accidents, road slides, and transportation issues is increasing day by day. Road damages, including potholes, cracks, and raveling, are the main causes of the above issues. If these damages are not identified and repaired at an early stage, they can lead to traffic accidents, vehicle damage, increased travel time, and higher maintenance costs. When we pay attention to the manual road development methods, they cause high costs, are time-consuming, and mainly depend on road development authorities. That means it fully relies on human judgment. When we consider all of these, it clearly shows the requirement for a reliable and efficient **Automated Road Condition Monitoring System**. In 2025, Sri Lanka recorded over 2,500 fatal road accidents causing around 2,700 deaths. Poor road conditions and delayed maintenance were major factors. This highlights the urgent need for an automated road condition monitoring system to detect damages early and improve safety.

To empower the proposed system, mobile phone cameras are used to capture road surface videos at a low cost, which serve as the image acquisition method. Image processing techniques provide an automated approach for analyzing road surface conditions. The captured video footage is divided into individual frames and processed to identify visual differences between damaged and undamaged road areas. **Image enhancement techniques are applied to improve the quality of the captured images by reducing noise and adjusting contrast, which helps to make road damage patterns more visible. Segmentation methods use features such as brightness changes, rough or uneven surface patterns, and clear boundaries to identify and extract damaged regions of the road surface.** By continuously analyzing video frames, the system enables consistent and objective road damage detection without relying on manual inspection.

The proposed Automated Road Condition Monitoring System is a practical and reliable way to find road surface damages early. It automatically detects damaged areas and checks their condition, helping road development authorities plan maintenance work better and prevent sudden road failures. For example, in Sri Lanka's rainy season, potholes and cracks can appear quickly on main and rural roads because of heavy rain and poor drainage. With this system, road surface videos taken by mobile cameras can be analyzed fast to spot damages before they get worse. This reduces the need for frequent manual inspections, saves time and money, and makes roads safer for drivers and pedestrians. Overall, the system helps make better decisions, lowers accident risks, and supports safer and more reliable transportation in Sri Lanka.